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Synthesis, electrochemical and photochemical properties of unsymmetrically substituted zinc phthalocyanine complexes

Pulane Matlaba, Tebello Nyokong*

Department of Chemistry, Rhodes University, Grahamstown 6140, South Africa

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Abstract

The synthesis of symmetrically and unsymmetrically substituted zinc phthalocyanine (ZnPc) derivatives are reported. The ZnPc derivatives are synthesized by ring expansion of subphthalocyanine complexes. Ring substitution is effected with *tert*-butyl phenol, naphthol and hydroxybenzoic acid. Comparison of the redox potentials for complexes substituted with varying numbers of *tert*-butyl phenol: 1 (complex **5**), 2 (complex **6**), 3 (complex **7**), 6 (complex **8**) and 8 (complex **9**) shows that the complexes with the highest number of substituents are more difficult to oxidize and easier to reduce. All the reported complexes are relatively photostable, with photobleaching quantum yields of the order of 10^{-5} . Singlet oxygen quantum yields for the complexes ranged from 0.22 to 0.68. © 2002 Elsevier Science Ltd. All rights reserved.

Keywords: Zinc phthalocyanine; Sensitizer; Singlet oxygen; Photobleaching

1. Introduction

Phthalocyanines are an interesting class of compounds with increasingly diverse industrial and biomedical applications including photosensitization [1], linear optics [2], catalysis [3], liquid crystals [4], and gas sensing [5]. Specificity in the applications of phthalocyanines can be introduced by modification of the phthalocyanine ring or changes in the central metal. Remarkable progress has been made in recent years in the use of phthalocyanine derivatives as sensitizers for photodynamic therapy (PDT) of cancer [6,7]. Zinc phthalocyanine complexes have good photosensitizing properties for PDT [8–10]. Sulfonated or carboxy zinc phthalocyanine complexes are water-soluble but form aggregates in solution [11–13]. Aggregation diminishes the photosensitizing ability of metallophthalocyanine (MPc) complexes. Amphiphilic unsymmetrically substituted phthalocyanines have been shown to have cell penetrating powers [14].

The major limitation of phthalocyanines for use in PDT is the lack of selectivity towards tumor cells. Unsymmetrically substituted phthalocyanines offer a possibility of improving selectivity since groups with different properties may be attached to the ring. We report in this work on the synthesis of unsymmetrically substituted zinc phthalocyanine complexes. Substituents, which enhance the solubility of the phthalocyanines, are employed. The most efficient method of synthesis of unsymmetrically substituted phthalocyanines is through synthesis of a symmetrically or unsymmetrically substituted sub-phthalocyanine (Sub-Pc), followed by ring enlargement using a substituted phthalonitrile [15–18].

It is widely believed that during PDT, the photosensitizer is excited to its triplet state, and then transfers the energy to the ground-state triplet oxygen $O_2(^3\Sigma_g)$ forming the excited-state singlet oxygen, $O_2(^1\Delta_g)$, through the so called Type II mechanism. The $O_2(^1\Delta_g)$ species is believed to be the chief agent in PDT. Type I mechanism involves electron transfer from the excited photosensitizer to the triplet state oxygen or to the biological substrate. Both Type I and Type II mechanisms may be involved during PDT [19,20]. It is important to determine singlet oxygen quantum yields of the photosensitizers. Studies on the photodegradation

* Corresponding author. Tel.: +27-46-603-8260; fax: +27-46-622-5109

E-mail address: t.nyokong@ru.ac.za (T. Nyokong).

(photobleaching) of the phthalocyanines are also important for determining the stability of the MPc complexes during PDT action. We report on the singlet oxygen quantum yields and the photobleaching studies of a series of unsymmetrically and symmetrically substituted zinc phthalocyanines (Scheme 1).

2. Experimental

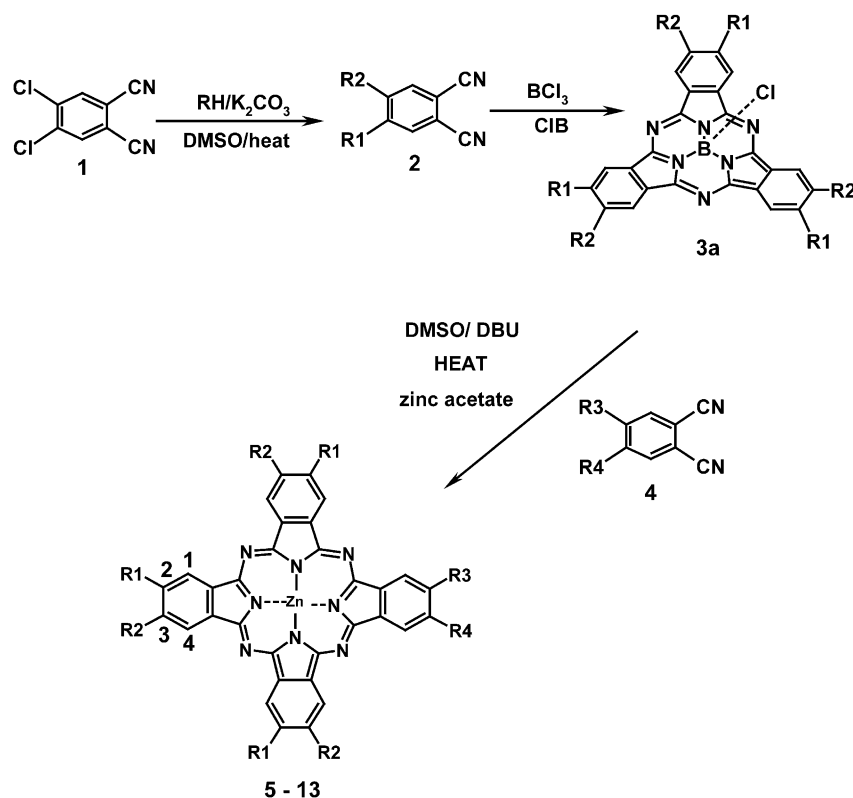
2.1. Materials

N,N'-dimethylformamide (DMF) used for electrochemical studies was freshly distilled. Dimethylsulfoxide (DMSO) was dried over alumina before use. Thionyl chloride, potassium carbonate, nitrophthalimide, dichlorophthalic anhydride, dicarboxynaphthalene, 2-naphthol, 4-*tert*-butylphenol, 4-hydroxybenzoic acid, 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU), diazabicyclooctane (DABCO), 1,3-diphenylisobenzofuran (DPBF) and chlorobenzene (CIB) were purchased either from Aldrich or Merck and were used as received. Tetrabutylammonium perchlorate (TBAP) was recrystallized from ethanol and used as an electrolyte for electrochemical studies.

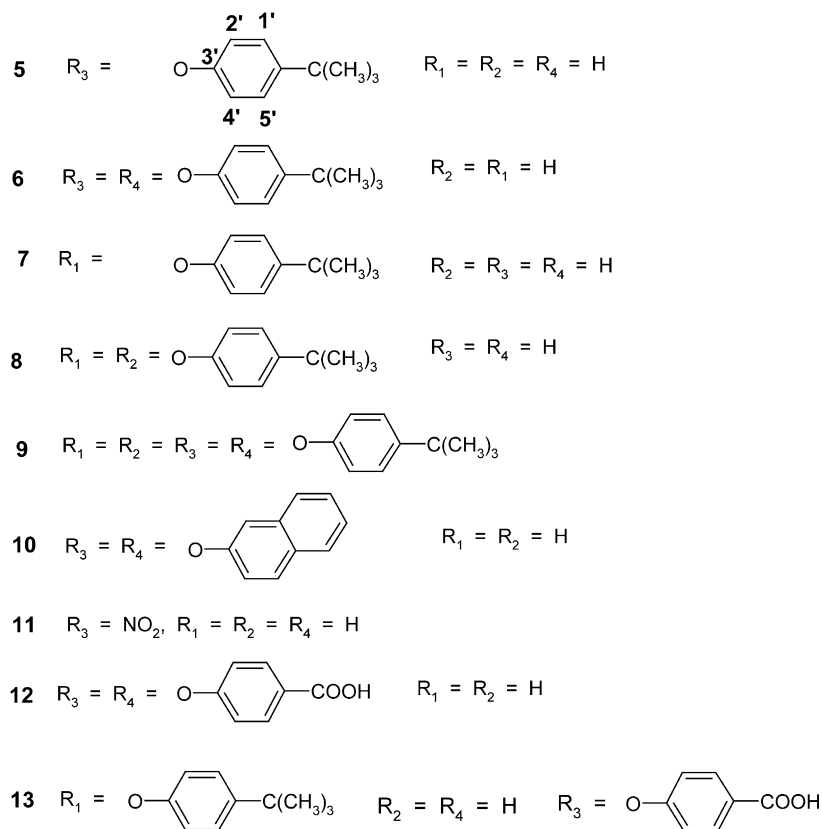
2.2. Synthesis

2.2.1. Substituted dicyanobenzene (2 and 4)

4,5-Dichlorodicyanobenzene (1) was synthesized from dichlorophthalic anhydride according to a reported procedure [21]. The disubstituted dicyanobenzenes ($R_1 = R_2$ (for 2) or $R_3 = R_4$ for 4) were then synthesized from 4,5-dichlorodicyanobenzene using the general procedure reported before [21] as follows: dry potassium carbonate (0.64 g, 0.0046 mol) was suspended in DMSO under nitrogen. To this suspension, 4,5-dichlorodicyanobenzene and the appropriate amount of the respective phenoxy substituent (2-naphthol, 4-*tert*-butylphenol or 4-hydroxybenzoic acid) were added. See Table 1 for the amounts of reagents used. Further aliquots of potassium carbonate were added after 4 h, and 24 h of stirring at room temperature (r.t.), giving total amounts shown in Table 1. The mixture was further stirred at r.t. and under nitrogen atmosphere for the total time shown in Table 1. The product was purified following literature methods [21]. Monosubstituted dicyanobenzenes ($R_2 = H$ (for 2) or $R_3 = H$ (for 4)) were synthesized by the procedure outlined above for the disubstituted derivatives using nitrophthalonitrile instead of dichlorodicyanobenzene. The amounts for the synthesis are shown in Table 1. 4-Nitrophthalonitrile was synthesized from 4-nitrophthalimide using reported procedures [22].



Scheme 1.



Scheme 1 (continued).

Dicyanonaphthalonitrile was synthesized from dicarboxynaphthalene following reported procedures [21].

2.2.2. Sub-phthalocyanines (**3a**)

The unsubstituted, tri- or hexa-substituted subphthalocyanines were synthesized by following reported procedures [15,16] as follows. The respective dichlorocyanobenzene (**1**), di- or mono-substituted dicyanobenzene (**2**, 0.4 g, 0.003 mol), was suspended in 1-chloronaphthalene (6 ml), stirred at r.t. in an atmosphere of nitrogen and in the presence of 0.001 mol (0.12 g) BCl_3 . The reaction mixture was heated to 220 °C for 0.5 h. The products were purified according to reference

[16], and characterized by UV–Vis, infrared and 1H NMR spectroscopies.

2.2.3. Sub-naphthalocyanine (Sub-NPc, **3b**)

Sub-naphthalocyanine was synthesized from dicyanonaphthalonitrile following the procedure outlined above for Sub-Pc, by replacing **2** with dicyanonaphthalonitrile.

2.2.4. Symmetrically (**9**) and unsymmetrically (**5–8, 10–14**) substituted zincphthalocyanines (Scheme 1)

These complexes were synthesized according to literature methods [23] by dissolving (0.25 g, 0.00092 mol) the unsubstituted or substituted dicyanobenzene (**4**) to

Table 1
Amounts of reagents used for synthesis of substituted dicyanobenzenes

Substituent	Amount of substituent added (g, mol)	4,5-Dichloro-dicyanobenzene (g, mol)	Total K_2CO_3 added (g, mol)	Total stirring time
<i>Di-substituted</i>				
4-Hydroxy benzoic acid	0.21, 0.00021	0.36, 0.0021	1.92, 0.0138	28 h
4- <i>tert</i> -Butylphenol	2.2, 0.015	1.0, 0.005	4.1, 0.12	28 h
2-Naphthol	8.6, 0.06	2.0, 0.0095	8.2, 0.24	28 h
<i>Mono-substituted</i>				
4-Dihydroxy benzoic acid	0.43, 0.0031	0.36, 0.0021	1.92, 0.0138	5 days
4- <i>tert</i> -Butylphenol	4.4, 0.03	3.4, 0.02	16.5, 0.12	48 h
2-Naphthol	1.3, 0.009	1.0, 0.0059	5.4, 0.039	5 days

be used for ring expansion in a mixture of freshly distilled DMSO–1-chloronaphthalene (4/2 ml) in the presence of DBU (0.1 g, 0.00065 mol). The reaction mixture was heated to 130 °C, then a suspension of the appropriate unsubstituted or substituted Sub-Pc (**3a**) or the sub-naphthacyanine (**3b**) (0.2 g, 0.00046 mol) and zinc(II) acetate dihydrate (0.11 g, 0.0005 mol) in the DMSO–1-chloronaphthalene solvent mixture was added drop-wise to the heated mixture over a period of 1 h. After cooling, glacial acetic acid (0.04 g, 0.65 mol) was added and the solvents evaporated. For complex **9**, there was no need for ring expansion of Sub-Pc, hence the complex was synthesized by reacting dicyanobenzene disubstituted with *tert*-butylphenol directly with zinc (II) acetate dihydrate. For all complexes, methanol was added and the product was separated by centrifugation. The product was further purified by thin layer chromatography (TLC), using ethyl acetate and toluene for elution.

2.2.5. 2-Mono(*tert*-butylphenoxy) phthalocyanato zinc (**5**)

Yield 35%. *Anal.* Calc. for $C_{42}H_{28}N_8O_1Zn \cdot 6(C_4H_8O_2)$: C, 63.16; H, 6.06; N, 8.93. Found: C, 63.99; H, 6.24; N, 8.14%.

IR (KBr), ν (cm^{-1}): 3962, 3056, 1715, 1602, 1558, 1507 (C–H); 1471, 1394, 1363, 1336, 1262 (C–O–C); 1235, 1174, 1091, 1046, 946, 803. 1H NMR (*d*-DMSO, 400 Hz): 8.90 (s, 8H, 1,4-Pc); 8.85 (s, 7H, 2,3-Pc); 8.10 (d, 2H, 1',5'-H); 7.40 (d, 2H, 2',4'-H); 1.30 (s, 9H, CH_3). λ_{max} (nm) (log ϵ) in DMF: 675 (5.09), 609 (3.71), 348 (3.96).

2.2.6. 2,3-Bis(*tert*-butylphenoxy) phthalocyanato zinc (**6**)

Yield 40%. *Anal.* Calc. for $C_{52}H_{40}N_8O_2Zn \cdot 6H_2O$: C, 63.58; H, 5.30; N, 11.41. Found: C, 61.54; H, 3.54; N, 13.41%. IR (KBr), ν (cm^{-1}): 3403, 2964 (C–H); 1601, 1508 (C–H); 1392, 1247, 1106, 994, 895 (C–O–C); 831, 745, 726, 477. 1H NMR (*d*-DMSO, 400 Hz): 9.20 (s, 6H, 1,4-Pc); 8.50 (s, 2H, 1,4-Pc); 8.17 (s, 6H, 2,3-Pc); 8.00 (d, 4H, 1',5'-H); 7.93 (d, 4H, 2',4'-H); 1.45 (s, 18H, CH_3). λ_{max} (nm) (log ϵ) in DMF: 678 (4.47), 611 (3.74), 359 (4.00).

2.2.7. 2,9,16-Tri(*tert*-butylphenoxy) phthalocyanato zinc (**7**)

Yield 43%. IR (KBr), ν (cm^{-1}): 3385, 2931 (C–H); 1709, 1625, 1485, 1331, 1250 (C–O–C); 1185, 1092, 887, 776, 751, 725, 467. 1H NMR (*d*-DMSO, 400 Hz): 9.48 (s, 8H, 1,4-Pc); 8.54 (m, 2H, 2,3-Pc); 8.10 (m, 3H, 2,3-Pc); 7.60 (d, 6H, 1', 5'-H); 7.30 (d, 6H, 2', 4'-H); 1.3 (s, 27H, CH_3). λ_{max} (nm) (log ϵ) in DMF: 669 (4.57), 604 (3.82), 335 (4.24).

2.2.8. 2,3,9,10,16,17-Hexa(*tert*-butylphenoxy) phthalocyanato zinc (**8**)

Yield 30%. IR (KBr), ν (cm^{-1}): 2946 (C–H); 1699, 1486, 1409, 1332, 1273 (C–O–C); 1091, 889, 774, 752, 467. 1H NMR (*d*-DMSO, 400 Hz): 9.80 (s, 8H, 1,4-Pc); 8.75 (s, 2H, 2,3-Pc); 8.00 (d, 12H, 1',5'-H); 7.70 (d, 12H, 2',4'-H); 1.3 (s, 54H, CH_3). λ_{max} (nm) (log ϵ) in DMF: 670 (4.75), 605 (3.91), 342 (4.09).

2.2.9. 2,3,9,10,16,17,23,24-Octa(*tert*-butylphenoxy) phthalocyanato zinc (**9**)

Yield 38%. IR (KBr), ν (cm^{-1}): 3056, 2959 (C–H); 2622, 2550, 2433, 1954, 1719, 1607, 1506, 1484, 1456, 1407, 1331, 1283 (C–O–C); 1244, 1209, 1164, 1114, 1092, 1059, 950, 886, 829, 794, 774, 751, 728, 634, 570, 499, 435. 1H NMR (*d*-DMSO, 400 Hz): 9.5 (s, 8H, 1,4-Pc); 7.5 (d, 16H, 1',5'-H); 7.0 (d, 16H, 2',4'-H); 1.2 (s, 72H, CH_3). λ_{max} (nm) (log ϵ) in DMF: 670 (4.73), 605 (4.20), 335 (4.48).

2.2.10. 2,3-Bis(naphthoxy) phthalocyanato zinc(II) (**10**)

Yield 20%. *Anal.* Calc. for $C_{42}H_{28}N_8O_1Zn \cdot 2(H_2O)$: C, 69.60; H, 3.57; N, 12.50. Found: C, 68.78; H, 2.70; N, 14.17%. IR (KBr), ν (cm^{-1}): 3053, 2930 (C–H); 2859, 2625, 2547, 1957, 1713, 1600, 1484, 1408, 1334, 1285 (C–O–C); 1210, 1165, 1059, 962, 913, 887, 809, 771, 752, 721, 634, 571, 500, 471, 433. 1H NMR (*d*-DMSO, 400 Hz): 9.25 (s, 8H, 1,4-Pc); 8.25 (s, 6H, 2,3-Pc); 8.8–7.5 (m, 14H, naphthoxy). λ_{max} (nm) (log ϵ) in DMF: 670 (4.97), 605(4.24), 335 (4.69).

2.2.11. 2-Mono-nitro phthalocyanato zinc(II) (**11**)

Yield 41%. *Anal.* Calc. for $C_{32}H_{15}N_9O_2Zn \cdot 2(C_4H_8O_2)$: C, 60.15; H, 3.88; N, 15.79. Found: C, 59.67; H, 3.18; N, 16.28%. IR (KBr), ν (cm^{-1}): 3931, 3332, 1950, 1702, 1652, 1458, 1322 (NO_2); 1162, 1055, 885, 724, 632, 567, 495, 433. 1H NMR (*d*-DMSO, 400 Hz): 9.70 (s, 8H, 1,4-Pc); 8.60 (s, 6H, 2,3-H). λ_{max} (nm) (log ϵ) in DMF: 670 (4.00), 605 (3.50), 343 (4.10).

2.2.12. 2,3-Bis(carboxyphenoxy) phthalocyanato zinc(II) (**12**)

Yield 48%. *Anal.* Calc. for $C_{46}H_{24}N_8O_6Zn \cdot 5H_2O$: C, 57.68; H, 3.76; N, 11.70. Found: C, 59.76; H, 3.58; N, 11.85%. IR (KBr), ν (cm^{-1}): 3437, 3050 (OH); 1602, 1502, 1403, 1271 (C–O–C); 1213, 1165, 1013, 951, 889, 858, 782, 745, 699, 618, 495, 433. 1H NMR (*d*-DMSO, 400 Hz): 9.4–9.6 (s, 8H, 1,4-Pc); 8.35 (d, 6H, 2,3-Pc); 7.95 (d, 4H, 1',5'-H); 7.50 (d, 4H, 2',4'-H); 2H from COOH are not seen as their peak is too weak compared with that of the Pc ring and the phenoxy ring. λ_{max} (nm) (log ϵ) in DMF: 671 (6.05), 606 (4.32), 357 (3.71), 284 (4.57).

2.2.13. 2,9,16-Tri(*tert*-butylphenoxy)-23-mono(carboxyphenoxy) phthalocyanato zinc(II) (13)

Yield 20%. *Anal.* Calc. for $C_{42}H_{28}N_8O_1Zn \cdot 2(H_2O)$: C, 66.19; H, 4.20; N, 14.71. Found: C, 66.57; H, 4.21; N, 16.36%. IR (KBr), ν (cm^{-1}): 3056, 2946 (C–H); 1716, 1658, 1597, 1499, 1453, 1270 (C–O–C); 1218, 1162, 1107, 1015, 883, 759, 690, 619, 486. 1H NMR (*d*-DMSO, 400 Hz): 9.40 (s, 8H, 1,4-Pc); 9.00 (s, 4H, 2,3-Pc); 7.55 (d, 8H, 2',4'-H); 7.42 (d, 8H, 1',5'-H); 1.28 (t, 27H, CH_3). λ_{max} (nm) (log ϵ) in DMF: 682 (4.97), 614 (4.24), 346 (4.69).

2.2.14. 2,3-Tribenzo phthalocyanato zinc(II) (14)

Yield 38%. *Anal.* Calc. for $C_{44}H_{22}N_8Zn \cdot 8H_2O$: C, 60.45; H, 4.35; N, 12.82. Found: C, 60.77; H, 5.67; N, 14.71%. IR (KBr), ν (cm^{-1}): 3389, 3180, 3056 (C–H); 1720, 1440, 1255, 1009, 797, 752, 615, 468. 1H NMR (*d*-DMSO, 400 Hz): 9.45 (d, 8H, 1,4-Pc); 8.30 (d, 6H, 6,7-H); 7.90 (d, 6H, 5,8-H). λ_{max} (nm) (log ϵ) in DMF: 672 (4.76), 607 (4.07), 344 (4.69).

2.3. Electrochemical methods

Electrochemical data were collected with the BioAnalytical Systems (BAS) model 100 B/W electrochemical workstation. For cyclic and square wave voltammetries, a platinum disk (3.0 mm diameter) was used as a working electrode. Platinum wire auxiliary and Ag wire pseudo-reference electrodes were employed for all electrochemical experiments. DMF containing ~ 0.5 mol dm^{-3} TBAP was employed as an electrolyte. Nitrogen gas was bubbled through the solutions prior to recording cyclic voltammograms, and a nitrogen atmosphere was maintained throughout the cyclic voltammogram scans. The ferrocinium–ferrocene ($Fc^+ - Fc$) couple was employed as an internal standard, and potentials were referenced versus saturated calomel electrode (SCE) using the reported [24] potential of 0.46 V (vs. SCE) for the $Fc^+ - Fc$ couple in DMF. The cathodic to anodic peak separation was found to be 80 mV for the $Fc^+ - Fc$ couple. Prior to cyclic voltammetry (CV) scans, the platinum working electrode was polished with alumina ($< 10 \mu m$) on a Buehler felt pad, followed by washing with deionized water and rinsing with methanol and DMF. For cyclic and square wave voltammetry studies concentrations were of the order of 10^{-4} mol dm^{-3} .

2.4. Photodegradation and determination of singlet oxygen studies

Photodegradation and singlet oxygen quantum yield determinations were carried out using experimental set-up which has been described in detail somewhere else [19,25]. Experiments were performed in air. Typically a 2 ml DMSO solution of the complex (about 1×10^{-5} mol

l^{-1}) was introduced to a 1 cm pathlength UV–Vis cell and photolyzed in the Q band region with a General Electric Quartz line lamp (300 W). For singlet oxygen quantum yield studies, photolysis was performed in the presence of DPBF as a singlet oxygen quencher. To avoid chain reactions induced by DPBF in the presence of singlet oxygen [26] the concentration of DPBF was lowered to $\sim 3 \times 10^{-5}$ mol l^{-1} . The DPBF decay at 417 nm was monitored. A 600 nm glass cut off filter (Schott) and a water filter were used to filter off ultraviolet and far infrared radiation. And an interference filter (Intor, 670 nm with a bandwidth of 20 nm) was placed in the light path before the sample. The light intensity at 670 nm was measured with a power meter (Lasermate) and was found to be 5×10^{16} photons $s^{-1} cm^{-2}$.

Quantum yields of singlet oxygen (Φ_Δ) photogeneration were determined using the relative method with unsubstituted zinc phthalocyanine (ZnPc) as reference, using the following relationship Eq. (1).

$$\Phi_\Delta = \Phi_\Delta^{PcZn} \frac{W \times I_{abs}^{PcZn}}{W^{PcZn} \times I_{abs}} \quad (1)$$

where, Φ_Δ^{PcZn} is the singlet oxygen quantum yield in the presence of unsubstituted ZnPc as a standard. For ZnPc standard, Φ_Δ^{PcZn} in DMSO is 0.67 [27]. W and W^{ZnPc} are the DPBF photobleaching rates in the presence of substituted ZnPc derivatives under investigation and unsubstituted ZnPc standard, respectively. I_{abs} and I_{abs}^{ZnPc} are the rates of light absorption by the substituted ZnPc derivatives and ZnPc standard, respectively. The initial DPBF concentrations were kept the same for both the ZnPc reference and the samples. No significant changes in the absorbance of the substituted ZnPc derivatives was observed during the singlet oxygen studies. All the experiments were carried out at r.t.

The quantum yields for photobleaching were determined using Eq. (2).

$$\Phi = -\frac{(C_t - C_0) \times V \times N_A}{I_{abs} \times S \times t} \quad (2)$$

where C_t and C_0 in mole l^{-1} are the concentrations of the substituted ZnPc derivatives after and prior to irradiation, respectively, the changes in concentration of the dyes was about 15% for these studies. Values of C_t and C_0 were obtained in DMF using the Q band extinction coefficient given in the Section 2. V is the reaction volume, S the irradiated area of the cell, t the irradiation time, N_A is the Avogadro's number and I_{abs} the overlap integral of the radiation source light intensity and the absorption of the substituted ZnPc derivatives (the action spectrum) in the region of the interference filter transmittance and is defined by Eq. (3).

$$I_{\text{abs}} = \int (1 - 10^{-A_{\lambda}}) I_{\lambda} d\lambda \quad (3)$$

where I_{λ} is the intensity of light and A_{λ} is the absorbance of the ZnPc derivative at wavelength, λ . Experiments were performed also whereby the solutions were deaerated with N_2 gas or saturated with O_2 in order to study the role of oxygen in the photodegradation process.

Triplet state life times were determined for complexes **5**, **7** and **8** using a flash photolysis system. The excitation pulse was provided by a Nd-Yag (in frequency-doubled mode, providing 160 mJ, 8 ns pulses of laser light at 0–10 Hz), pumping a Lambda-Physik FL2002 dye laser. Single laser pulse energies used ranged from 0.001 to 2 mJ. A 75 W xenon arc lamp provided the analyzing light. Triplet state quantum yields (ϕ_T) were calculated by a comparative method using zinc phthalocyanine as a standard (ϕ_T ZnPc = 0.25 [28]).

UV–Vis spectra were recorded on a Varian 500 UV–Vis–NIR spectrophotometer. IR spectra (KBr pellets) were recorded on a Perkin–Elmer Spectrum 2000 FTIR spectrometer. ^1H nuclear magnetic resonance (NMR, 400 MHz) spectra were obtained in CDCl_3 and d_6 -DMSO using the Bruker EMX 400 NMR spectrometer.

3. Results and discussion

3.1. Synthesis and characterization

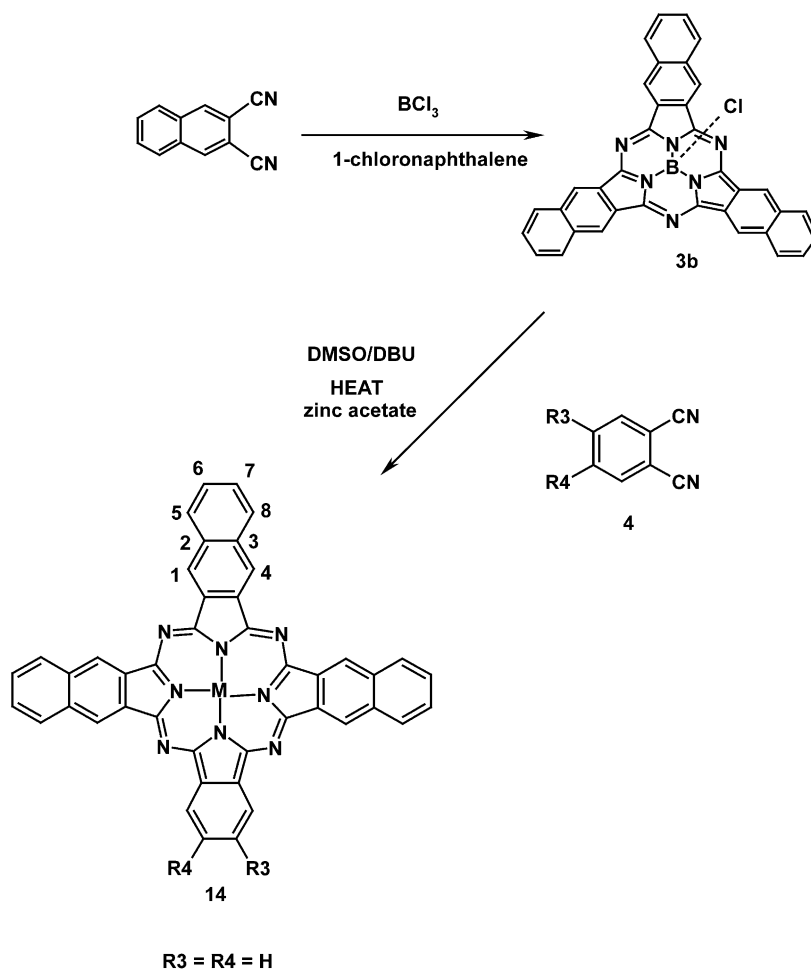
Synthesis of unsubstituted and substituted Sub-Pcs has been reported by several researchers [15,16,29–31]. The Sub-Pc complexes have been employed as starting materials for the formation of unsymmetrically substituted MPc complexes by ring expansion using substituted or unsubstituted phthalonitriles. The use of Sub-Pcs as intermediates for the synthesis of unsymmetrically substituted phthalocyanines has gained popularity over using statistical condensation of two differently substituted phthalonitrile derivatives. The later results in a complex mixture of products. However, using the reaction mixture containing phthalonitriles and boron trihalide, gives ring-halogenated Sub-Pc, in addition to the desired products. The halogenated Sub-Pcs is difficult to separate. The Sub-Pc complexes were used to prepare the unsymmetrically substituted zinc phthalocyanines according to Schemes 1 and 2. The product mixture will inevitably contain halogenated Pc derivatives and the desired products were separated using preparative TLC.

The ZnPc derivatives (**5**, **6**, **10–12**) were synthesized using unsubstituted Sub-Pc and ring expanding using dicyanobenzene which was mono-substituted with *tert*-butylphenol (**5**) or nitro (**11**); and dicyanobenzene disubstituted with *tert*-butylphenol (**6**), naphthol (**10**) or

hydroxybenzoic acid (**12**). Complexes **7**, and **13** were synthesized using Sub-Pc substituted with three *tert*-butylphenol groups and ring expanding with unsubstituted dicyanobenzene (**7**) or hydroxybenzoic acid (**13**). Complex **8** was synthesized using Sub-Pc substituted with six *tert*-butylphenol groups and ring expanding with unsubstituted dicyanobenzene (**8**). Complex **9** was synthesized from dicyanobenzene disubstituted with *tert*-butylphenol. Complex **14** was synthesized from sub-naphthalocyanine (Sub-NPc, **3b**, Scheme 2) and ring expanding with dicyanobenzene (**4**). In all cases an excess of the dicyanobenzene (over the Sub-Pc or Sub-NPc) was employed. The complexes gave satisfactory spectroscopic analyses, confirming the absence of chlorinated derivatives. The formation of the ZnPc derivatives was monitored using UV–Vis spectroscopy, Fig. 1. An expected decrease in the absorption band of the Sub-Pc in the 570 nm region and the simultaneous formation of the Q and B bands of the phthalocyanine was observed. The reaction was complete within 1–4 h depending on the complex. Complex **5** required only 1 h, while the rest of the complexes needed 2–4 h. Reaction times much longer than these have been reported for these types of reactions [29]. The lack of clear isosbestic points in Fig. 1, could be due to the presence of differently ring substituted isomers.

Yields of the ZnPc derivatives were comparable or higher than those reported in the literature [29] for synthesis of Pcs from Sub-Pcs. The complexes were soluble in DMF or DMSO to a varying degree. Complexes containing *tert*-butylphenol substituents or hydroxybenzoic acid were found to be more soluble. Complexes **1–9**, **12** and **13** were soluble in chloroform. Elemental analysis was not obtained for some of the complexes because of the difficulty in obtaining reliable results due to the presence of differently ring substituted isomers.

In general the spectra of the complexes was typical of non-aggregated species at concentrations lower than $1 \times 10^{-5} \text{ mol dm}^{-3}$, as judged the fact that Beer's law was obeyed, Fig. 2. The Q band wavelengths of complexes under discussion ranged from 669 (complex **7**) to 682 nm (complex **13**) in DMF, Table 2. The Q band maximum for complex **13** is shifted to longer wavelengths compared with the rest of the complexes. This complex contains two different type of substituents: *tert*-butylphenol and hydroxybenzoic acid on the phthalocyanine ring, while the rest of the complexes contain one type of substituent. It is well known [30] that the Q band shifts to longer wavelengths with enlargement of π -conjugated system and that the shifts became less, the larger the molecules. On going from complex **5–9**, the bulkiness of the molecule increases as the number of *tert*-butylphenol groups is increased from 1 (**5**) to 8 (for **9**). For these complexes, the wavelength of the Q band increases as follows: **6** (678) > **5** (675) > **8** (670) = **9**



Scheme 2.

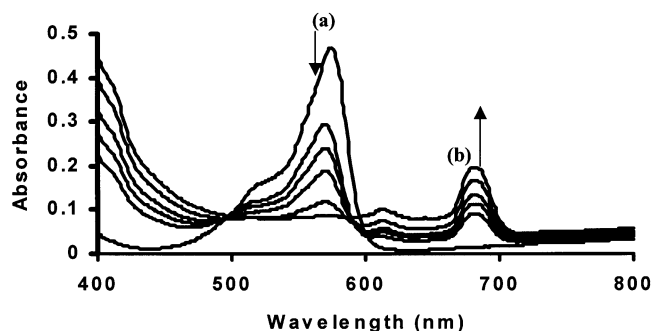


Fig. 1. Changes in UV–Vis spectra during the formation of ZnPc derivatives from Sub-Pc. Spectra at the start (a), and at the end (b), of the reaction. Solvent, DMF.

(670) ~ 7 (669 nm). Thus the Q band wavelength is shorter for the more bulky molecules (complexes 7, 8 and 9) and longer for less bulky molecules (complexes 5 and 6), in agreement with the literature [30]. The shift to longer wavelength could be due to either the destabilization of the highest occupied molecular orbital (HOMO) or the stabilization of the lowest unoccupied molecular orbital (LUMO). It has been observed that the shifting of the Q band observed on increasing the π -system, is

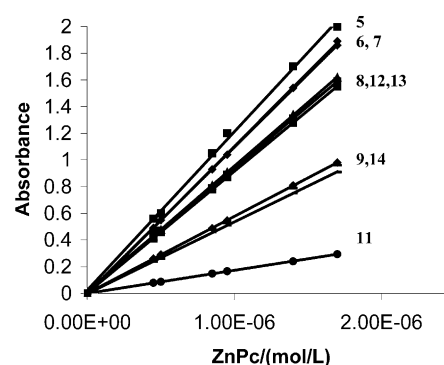


Fig. 2. Beer's law dependence of the ZnPc derivatives in DMF.

due mainly to the destabilization of the HOMO. Lack of symmetry in phthalocyanines results in the splitting in the Q band [29]. Fig. 3 shows a representative spectra of the complexes, and no split in the Q band was observed showing that the complexes still have some symmetry.

Infrared spectra observed for the phthalocyanine complexes are all very similar. All the compounds with the exception of 11 and 14, show a band at $\sim 1200\text{ cm}^{-1}$ (C–O–C vibration) which is characteristic of a phenyl ether functionality.

Table 2

Quantum yields for singlet oxygen and photobleaching for the ZnPc derivatives in DMSO

Complex	λ_Q band (nm)	Φ_{Δ}^a	$10^5 \Phi_{\text{photobleaching}}$
5	675	0.68 (114)	3.9
6	678	0.49	4.1
7	669	0.57 (240)	5.8
8	670	0.51 (129)	6.5
9	670	0.52	3.3
10	670	0.43	4.1
11	670	0.49	3.8
12	671	0.40	3.7
13	682	0.22	3.3
14	672	0.32	3.0

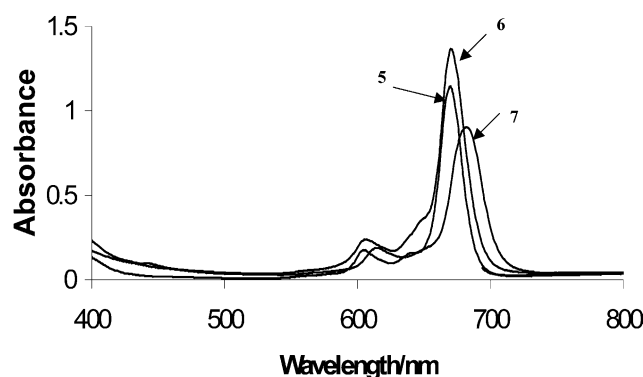


Fig. 3. Electronic absorption spectra of selected complexes (5–7) in DMF. Concentration = $5 \times 10^{-6} \text{ mol dm}^{-3}$.

^1H NMR spectra of the complexes was recorded in *d*-DMSO. Characteristic resonances due to the peripheral (1,4 positions, see Schemes 1 and 2 for numbering) protons of the phthalocyanes were observed as mainly singlets between 8.5 and 9.8 ppm, integrating for a total of eight protons. A doublet was observed for these protons for complex 14. In addition, 2,3 protons for the phthalocyanine ring were observed at resonances ranging from 8.0 to 9.0. For mono-substituted ZnPc derivatives (5 and 11) a singlet integrating for a total of seven protons (2,3 protons) was observed for each complex. For di-substituted ZnPc derivatives (6, 10 and 12), singlets or doublets integrating for six protons was observed for each complex. Tri-substituted complex (7) gave a multiplets integrating for a total of five protons, and for the hexasubstituted complex (8), a singlet integrating for two protons was observed as expected. Tetrasubstituted complex 13 gave a singlet for the 2,3 protons of the phthalocyanine ring, integrating for a total of four protons. Complex 9 is octasubstituted hence contained no 2,3 protons.

The substituents on the phthalocyanine ring showed resonances due to the aromatic rings and these integrated correctly. In the Section 2, the ring protons of the substituents are denoted as 2',4'-H or 1',5'-H, and the numbering is shown for complex 5 in Scheme 1.

3.2. Voltammetric studies

The redox properties of the ZnPc derivatives were studied using both CV and Osteryoung square wave voltammetry (OSWV) in DMF containing TBAP. Figs. 4 and 5 show the cyclic voltammogram and square wave voltammograms for complexes 8 and 5, respectively. Two reduction couples labeled II and III was observed, and oxidation couple(s) (labeled as I or Ia and Ib) were also observed. The cyclic voltammograms for the first scan were similar to those of the second and subsequent scans, suggesting that no complicated reactions occurs following oxidation or reduction. For all couples, the cathodic to anodic peak currents (i_p/i_{p_a}) were near unity, and cathodic to anodic peak separation ranged from 80 to 150 mV, suggesting reversible to quasi-reversible behavior. The cathodic to anodic peak separation was found to be 80 mV for the $\text{Fc}^+ - \text{Fc}$ couple. The peak currents increased linearly with the square root of scan rates, for scan rates ranging from 5 to 800 mV s^{-1} , indicating that the electrode reactions are purely diffusion-controlled. The broadness of some of the redox couples could be a result of the presence of isomers.

All processes are ring based since the central Zn metal is electroinactive [32]. Using controlled potential coulometry, couples II and III are assigned to ring reduction in ZnPc derivatives and the formation of $\text{Pc}^{-3} - \text{Pc}^{-2}$ and $\text{Pc}^{-4} - \text{Pc}^{-3}$, respectively. The separation between the first and second ring reductions was found to range between 0.4 and 0.5 V in agreement with the separation reported [32] for ring based reductions in non-transition metal phthalocyanines. For complex 5, Fig. 5, two closely spaced couples (Ia and Ib) were observed for the oxidation process. For complex 8, the splitting was more easily observed in the SWV, Fig. 4. The splitting

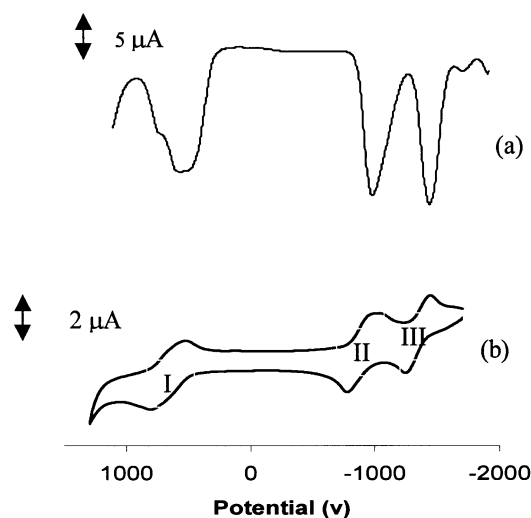


Fig. 4. Osteryoung square wave voltammogram (a) and cyclic voltammogram (b) of $1 \times 10^{-4} \text{ mol dm}^{-3}$ of complex 8 in DMF containing 0.1 mol dm^{-3} TBAP. Scan rate = 100 mV s^{-1} .

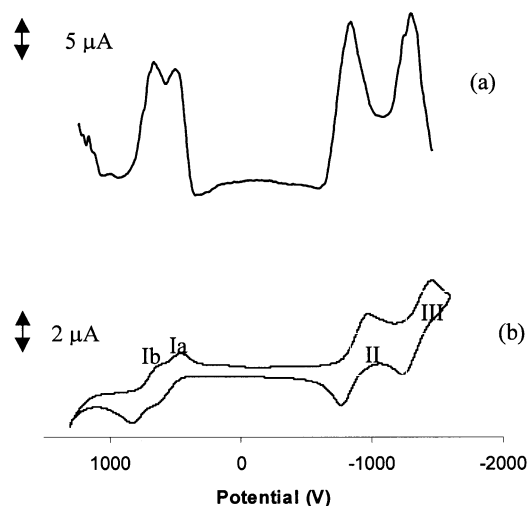


Fig. 5. Osteryoung square wave voltammogram (a) and cyclic voltammogram (b) of 1×10^{-4} mol dm $^{-3}$ of complex **5** in DMF containing 0.1 mol dm $^{-3}$ TBAP. Scan rate = 100 mV s $^{-1}$.

was observed to varying extents for the ZnPc derivatives. The observation of two closely spaced oxidation peaks has been attributed to the presence of both the aggregated and unaggregated substituted ZnPc complexes in solution [32]. At the concentrations employed for voltammetric studies in this work ($\sim 10^{-4}$ mol dm $^{-3}$), aggregation of the ZnPc derivatives is expected and was confirmed by the broadening of the spectra at these concentrations. The observation of two closely spaced oxidation peaks could also be a result of the presence of isomers. Thus the two couples observed for the oxidation process in Fig. 5, are attributed to the ring oxidation of the aggregated and non-aggregated species. Table 3 compares the half-wave potentials ($E_{1/2}$) for complexes **5**, **7**, **8** and **9** containing increasing number of *tert*-butyl phenol substituents. The Pc ring becomes easier to reduce and more difficult to oxidize as the number of *tert*-butyl phenol substituents increase from 1 (complex **5**) to 8 (complex **9**), Table 3. The presence of electron donating methyl groups is expected to make the ring more difficult to reduce and easier to oxidize. Thus it is expected that as the number of *tert*-butyl phenol substituents increases, the ring should be more difficult to reduce and easier to oxidize. The reverse is observed in Table 3.

Table 3
Comparison of redox potentials for complexes **5**, **7**, **8** and **9** substituted with varying numbers of *tert*-butyl phenol substituents

Complex	Number of <i>tert</i> -butyl phenol substituents	$E_{1/2}$ (V vs. SCE)		
		I Pc $^{-1}$ -Pc $^{-2}$	II Pc $^{-2}$ /Pc $^{-3}$	III Pc $^{-3}$ -Pc $^{-4}$
5	1	0.35, 0.55	−1.08	−1.56
7	3	0.75	−0.82	−1.32
8	6	0.70	−0.84	−1.32
9	8	0.97	−0.68	−1.12

Solvent, DMF containing TBAP. Pc(−2), phthalocyanine dianion.

3.3. Photobleaching and singlet oxygen studies

Photobleaching of complexes is characterised by complete disappearance of the spectra due to the degradation of the phthalocyanine ring. Photobleaching studies were undertaken in order to determine the effects of the different substituents on the stability of the ZnPc derivatives in the presence of light.

Photobleaching studies were performed in DMSO and the quantum yields for photobleaching are listed in Table 2. Photobleaching quantum yields were of the order of 10^{-5} for all the complexes. Photodecomposition of phthalocyanine complexes usually occurs through oxidative cleavage [33]. Comparing complexes which contain an increasing number of the *tert*-butyl phenol substituents (**5**–**9**), we observe a low photobleaching quantum yield (hence higher stability) for complex **9** which was shown to be more difficult to oxidize by electrochemical methods, hence confirming that oxidative degradation occurs for these complexes. Complex **5** was easier to oxidize electrochemically, but is relatively stable, suggesting that it is not only the oxidative mechanism which plays a part in the degradation of the ring in these ZnPc derivatives. It is normally accepted [33] that during photodegradation of MPc complexes O $_2$ ($^1\Delta_g$) is generated which attacks the phthalocyanine ring. However, the attack of O $_2$ ($^1\Delta_g$) on the Pc ring may also be affected by the shielding effects of the ring substituents [34]. It has been shown [33,34] that the singlet oxygen interacts with the phthalocyanine, resulting in the macrocycle destruction and formation of phthalimide as the product of photodegradation.

Quantum yields of photobleaching increased when oxygen was bubbled to solutions, confirming that oxidative degradation occurs. Inhibition of the photobleaching process in the presence of the singlet oxygen quencher, DABCO, Fig. 6 is in agreement with the involvement of singlet oxygen in the degradation of the phthalocyanine macrocycle.

Quantum yields of singlet oxygen photogeneration (Φ_Δ) of compounds **5**–**14** are listed in Table 2. These values range from 0.22 to 0.68. No photobleaching of complexes was observed during the determination of singlet oxygen quantum yields, since these processes

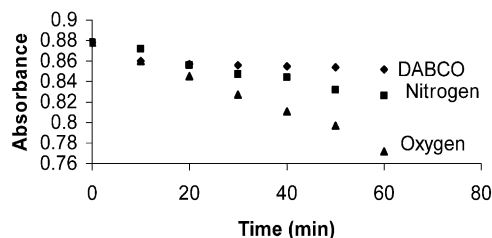


Fig. 6. Kinetic curves for the photobleaching of complex **5** in DMSO saturated with oxygen and nitrogen, and containing DABCO ($2 \times 10^{-3} \text{ mol dm}^{-3}$).

occur much slower than singlet oxygen production. It is well known that aggregation of the phthalocyanine complexes results in decreased photochemical activity due to enhanced probability of radiationless decay of excited states. The concentration employed for Φ_{Δ} and photobleaching studies was $1 \times 10^{-6} \text{ mol dm}^{-3}$, hence no aggregation is expected according to Fig. 2. Comparing complexes which contain an increasing number of the *tert*-butyl phenol substituents (**5**–**9**), complex **5** with one substituent gave the largest value for Φ_{Δ} . Triplet life times (τ_T) were determined for complexes **5**, **7** and **8** and are listed in Table 3. Of the three complexes, the largest τ_T value was observed for complex **7** and the lowest for **5** even though the latter has the highest Φ_{Δ} value. A direct correlation between Φ_{Δ} and τ_T or Φ_{Δ} was also not evident for substituted MPC complexes reported in the literature [3]. The lowest Φ_{Δ} value was observed for complex **14** containing naphthalene units.

In conclusion voltammetric behavior of the substituted ZnPc derivatives under discussion show that as the number of *tert*-butyl substituents increase, the complexes become more difficult to oxidize and easier to reduce. The complexes are relatively stable to photo-degradation with quantum yields of photobleaching of the order of 10^{-5} . The majority of the complexes have singlet oxygen quantum yields in the 0.4 and 0.5 region, hence show promise as candidates for PDT.

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